

The Compute Merit Order: A Negative Result

Electricity cost is roughly one per cent of GPU rental revenue, and the geographic curtailment sequence it implies is not binding

Compute Merit Order Project

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Abstract

A GPU cluster earns a compute price per GPU-hour and pays an electricity cost per GPU-hour. The difference is a spread superficially analogous to the spark spread in power markets, and it implies a shutdown price below which a fleet is uneconomic to run. Because regional power costs differ, shutdown prices differ, which is said to imply a global merit order in which compute capacity curtails in a predictable geographic sequence. We construct that merit order from open data for seven European bidding zones and find that **it exists, binds in roughly two hundredths of one per cent of hours, and is worth about a dollar per GPU per year**. Evaluated at hourly resolution — the resolution at which a curtailment decision is actually taken — the right to stop running is an hourly option on the spread. Its intrinsic value is 1.30 USD per GPU-year in Belgium, 0.67 USD in the Netherlands, 0.12 USD in Germany/Luxembourg and exactly zero in the other four zones, and it is non-zero only at the cheapest of the nine revenue bases we measure. Against roughly 7000 USD per GPU-year of revenue that is about 0.02 %. We stress that an earlier version of this analysis evaluated the same data against *annual mean* prices and concluded that no zone is ever uneconomic; that conclusion was an artefact of aggregation, and Belgium’s dearest hour exceeds its own break-even price by 56 %. Power ranges from 1.0 % of revenue at H100 on-demand list prices to 16.5 % at A100 spot, so the answer depends far more on which revenue basis is chosen than on anything in the cost model. Cross-zone dispersion in shutdown prices is 1.65x on our published sample, compressing to 1.19x once non-power operating cost is added — an algebraic property, not a sample artefact. We recover Azure’s 1, 3 and 5-year committed rates from published reservation data, a discount to list routinely described as confidential. The spark-spread analogy survives only in a weak form: the mechanism is real and priceable, but at current prices it is worth two basis points of revenue, and the economics that matter for AI infrastructure lie in capital recovery and siting rather than in dispatch.

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1 Introduction

Spark spreads — the difference between the wholesale price of electricity and the fuel cost of generating it — are a standard tool in power-market economics, used to decide when a gas-fired plant is worth dispatching. An analogous quantity is often asserted to govern AI compute: a GPU cluster sells compute at some price per GPU-hour, its main variable cost is electricity, and when the compute price falls below that cost plus non-power operating expense, a rational operator curtails. Because industrial power costs vary substantially across regions, the shutdown prices implied by this logic vary too, which would produce a global merit order — a predictable geographic sequence in which compute capacity curtails as compute prices fall.

This paper tests that proposition and reports a negative result. The mechanism is real and we compute it explicitly, but its magnitude is far too small for it to bind at observed prices. The reason is a simple quantitative asymmetry that the analogy obscures: for a combined-cycle gas turbine, fuel is typically 60 % to 90 % of short-run marginal cost, so a change in gas prices moves the dispatch decision directly. For a GPU fleet renting at on-demand list prices, electricity is of order 1 % of revenue, rising to 17 % only at the cheapest spot price we observe. A threefold difference in a one-per-cent cost item is a two-percentage-point difference in margin, not a dispatch signal.

We test two claims:

1. **The compute merit order.** That regional power-cost differences produce a geographic curtailment sequence as compute prices fall (Section 5). We test curtailment, not retirement: the two are distinct decisions with distinct cost bases, and conflating them is a category error we discuss in Section 3.
2. **Chips versus power.** That the binding constraint on AI buildout is grid interconnection and power availability rather than chip supply (Section 6).

We are able to test the first and report it as not binding. We are *not* able to test the second with the data assembled so far, and we say so rather than gesture at it (Section 6).

A note on timing. We do not claim, as earlier drafts of this project did, that comparable compute-price data first appeared in 2026: at least one commercial index has published weekly observations since July 2024 (`docs/SOURCES.md`), and our own sources file contradicted that earlier claim. What is new here is not the existence of such data but the construction of a *freely redistributable* series from vendor-published prices, which is what makes an independently auditable spread computable at all.

2 Data and methods

2.1 Free-data constraint

This project publishes its dataset under CC-BY-4.0 and holds itself to a hard constraint: **a source is eligible only if it is free to access and permits redistribution of derived values**. A number a reader cannot fetch and re-derive is not auditable, and a number we are contractually barred from republishing cannot enter `/data/final` at all. This ruled out the two credible commercial compute-price indices (Section 8) and, as described below, five of the twelve power zones we collected.

2.2 Power prices

Day-ahead prices come from the Energy-Charts API operated by Fraunhofer ISE (Fraunhofer ISE, 2026), which requires no authentication. We collected 2026 year-to-date hourly prices for twelve European bidding zones. Energy-Charts returns a licence *per zone*, and they differ: seven zones arrive as CC-BY-4.0 via Bundesnetzagentur/SMARD, while five exchange-sourced zones (Spain, Portugal, Finland, Sweden SE3, Italy North) carry “for private and internal use only”. Those five are fetched and analysable locally but their derived values are withheld from the published dataset; the omission is recorded in `docs/WITHHELD_ZONES.md` rather than left silent. All results below therefore rest on seven zones, which is a materially smaller sample than we set out to use.

Prices are converted from EUR/MWh to USD kW⁻¹ h at the ECB monthly-average reference rate (European Central Bank, 2026), mean 1.163 365 over the period.

2.3 Compute prices

Because no free redistributable compute-price index exists (Section 8), we construct one from published cloud list prices, which are facts about public offers and can be recorded with attribution. We query the Azure Retail Prices API (Microsoft, 2026b), which is free and unauthenticated, for a fixed basket of GPU SKUs across three regions. Accelerator counts per SKU — required to convert a price per instance-hour into a price per GPU-hour — are read from Microsoft’s public documentation repository (Microsoft, 2026a) and recorded per SKU with their own source URL, never inferred from SKU naming.

We filter to Linux rows. Windows rows are excluded because they embed OS licensing, which is not a cost of running an accelerator, and expired price rows are excluded because they are history.

We publish five price bases: on-demand list, spot, and 1, 3 and 5-year reserved. The reserved series matters. Committed-rate pricing is widely described — including in earlier drafts of this paper — as confidential and unobtainable. It is not: Azure publishes reservation totals through the same free API, and dividing by the term’s hours yields a comparable hourly rate. The 5-year H100 rate is 60 % below list. Any analysis using list price as the revenue of a committed operator overstates it by roughly 2.5x, and we previously made exactly that error.

Accelerator	Price basis	USD per GPU-hour	% of list
H100	On-demand list	14.13	100
H100	Reserved 1-year	9.05	64
H100	Reserved 3-year	6.20	44
H100	Reserved 5-year	5.65	40
H100	Spot	2.61	18
A100	On-demand list	4.32	100
A100	Spot	0.82	19

Table 1: Published Azure price bases, mean across sampled regions. The choice of basis moves the revenue leg by 5x and is therefore a larger modelling decision than anything in the cost model.

2.4 Cost model

Delivered power per accelerator-hour is

$$\text{delivered_kW} = (\text{TDP} + \text{overhead}) \times \text{PUE} = (0.700 + 0.180) \times 1.25 = 1.10 \text{ kW}, \quad (1)$$

held as a single named constant and swept over 0.9 kW to 1.4 kW in Section 7. We flag that the 0.180 kW overhead term is a residual chosen to reconcile the other two figures with a 1.10 kW total, not an independent measurement.

3 The compute spark spread

Power cost per GPU-hour is

$$\text{power_cost} = \text{delivered_kW} \times \text{price}_{\text{USD/kWh}}. \quad (2)$$

The shutdown price adds non-power marginal operating expense:

$$\text{shutdown_price} = \text{power_cost} + \text{opex}_{\text{marginal}}. \quad (3)$$

Headroom against a compute price is

$$\text{headroom} = \frac{\text{compute_price} - \text{shutdown_price}}{\text{compute_price}}. \quad (4)$$

What this cost is, and is not. For a fleet whose accelerators are already purchased, capital cost is sunk and therefore irrelevant to the decision to keep running: the operator curtails when revenue falls below *avoidable* cost. Equation (3) is that avoidable cost, and it is the correct quantity for a shutdown question. It is emphatically not the right quantity for an investment question. A fleet can sit far above its shutdown price while being a value-destroying investment, because capital recovery dominates its full cost. We compute only the former, and Section 9 argues the latter is where the economically interesting question actually lives.

We do not assign a point estimate to $\text{opex}_{\text{marginal}}$. No public source for it was identified, so it is swept over 0.00 USD to 0.20 USD per GPU-hour and every result below is reported across that sweep.

4 Regional cost surfaces

4.1 Power

Across the seven publishable zones, 2026 year-to-date mean day-ahead prices run from 0.0847 USD kW⁻¹ h (France) to 0.1296 USD kW⁻¹ h (Austria), a spread of 1.53x. Negative price hours are common, totalling 1823 across the sample, which matters because it means the power cost of compute is occasionally negative and the shutdown constraint cannot bind at all in those hours.

The hub taxonomy does not predict cost. The project’s own supply-rich versus constrained classification fails on this sample. The cheapest zone is France, classified constrained. Norway NO2, classified supply-rich, is the second dearest on annual means at 0.1173 USD kW⁻¹ h — it is the southern Norwegian zone, heavily interconnected to Germany and Great Britain, and therefore prices toward its neighbours rather than toward Nordic hydro. Any hub-level narrative that assumes “Nordics are cheap” without checking the specific bidding zone is wrong, and the geographic intuition behind the merit-order thesis is weaker than it appears.

4.2 Compute

As of 2026-08-03, Azure’s published on-demand list price averages 14.13 USD per GPU-hour for H100 and 4.32 USD per GPU-hour for A100 across the regions sampled (Figure 1). Spot prices for the same silicon clear at roughly a fifth of list, so the dispersion *within* the compute price exceeds the dispersion in power costs that motivates this paper.

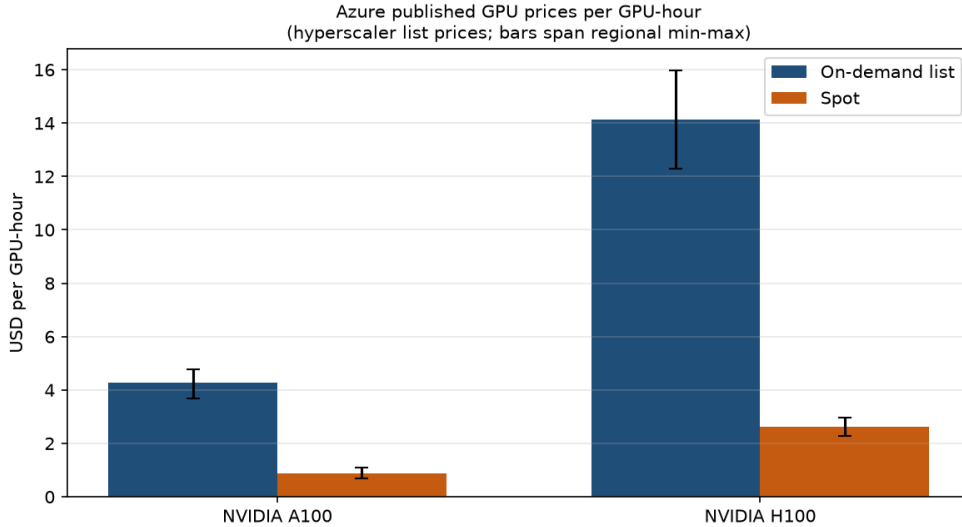


Figure 1: Published Azure GPU prices per GPU-hour. Bars span the regional minimum to maximum. On-demand list price is an upper bound on realised revenue; spot is shown for contrast, not as a revenue proxy.

5 The merit order

Figure 2 is the central result. Panel (a) is the merit order itself: zones sorted ascending by avoidable cost per GPU-hour. Panel (b) places that ladder on the same logarithmic axis as observed compute prices, which is the only way to show both on one chart — and the gap between them is the finding.

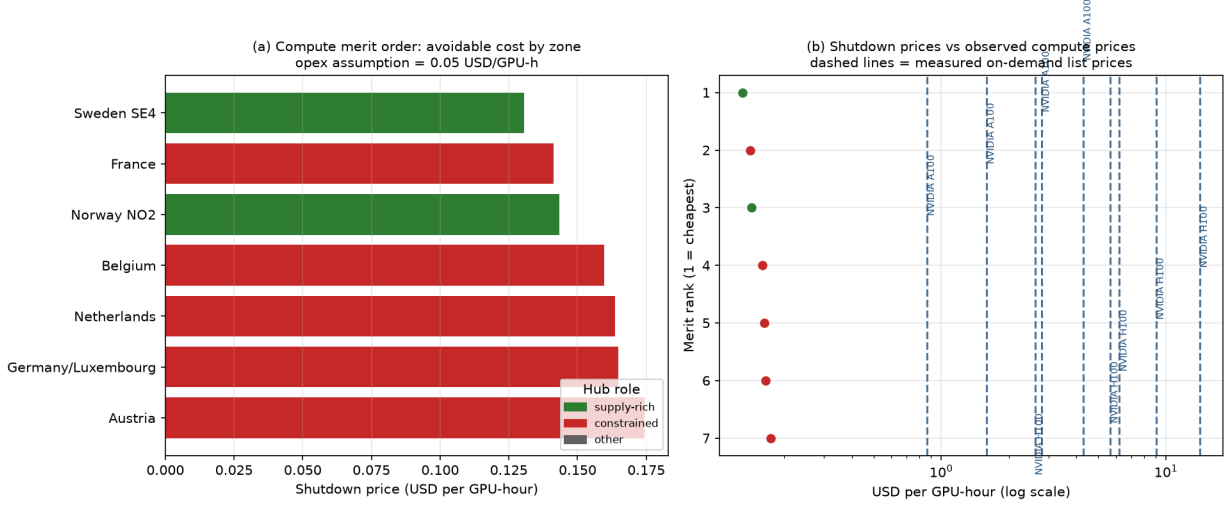


Figure 2: The compute merit order. (a) Avoidable cost by zone at the midpoint opex assumption. (b) The same shutdown prices against measured on-demand compute prices, log scale. The ordering is real; the gap is two orders of magnitude.

5.1 The order exists

The curtailment sequence is well defined. At the midpoint opex assumption and H100 prices, zones would go uneconomic in the order Austria, Norway NO2, Germany/Luxembourg, Belgium, Netherlands, Sweden SE4, France. So the qualitative claim — that a geographic sequence exists and is computable from public data — survives.

5.2 Resolution matters more than anything else in this model

An earlier version of this analysis compared each zone’s *annual mean* power price against the compute price, and concluded that no zone is ever uneconomic. That conclusion was false, and the way it was false is the most useful methodological result in this paper.

Curtailment is an hourly decision. An operator compares this settlement period’s avoidable cost against this period’s revenue, and a spark spread is evaluated per period for exactly that reason. Collapsing 36 281 zone-hours into seven annual means destroys the upper tail, and the upper tail is the entire phenomenon. Belgium’s mean power price is 0.114 USD kW⁻¹ h; its dearest hour is 1.086 USD kW⁻¹ h, nearly ten times higher and 56 % above its own break-even threshold. Aggregation, not economics, produced the null.

5.3 The right to curtail, priced

Evaluated hourly, the right to stop running when the spread is negative is a strip of hourly options on that spread, with strike equal to the compute price. Its intrinsic value over the sample is

$$V = \sum_t \max(0, \text{avoidable_cost}_t - \text{compute_price}), \quad (5)$$

the loss avoided by curtailing rather than running regardless. This is intrinsic value only — no volatility model, no discounting — and is therefore a lower bound.

Zone	Role	Uneconomic h/yr	Option value (USD/GPU-yr)
Belgium	constrained	5.07	1.30
Netherlands	constrained	3.38	0.67
Germany/Luxembourg	constrained	3.38	0.12
Austria	constrained	0.00	0.00
Sweden SE4	supply-rich	0.00	0.00
France	constrained	0.00	0.00
Norway NO2	supply-rich	0.00	0.00

Table 2: Intrinsic value of the right to curtail, at the A100 spot price (0.86 USD/GPU-h) and 0.05 USD/GPU-h non-power opex — the only one of our nine revenue bases at which the option is ever in the money. Annualised from a partial-year sample.

So the merit order is worth roughly 1.30 USD per GPU-year in Belgium, and nothing at all in four of seven zones. Against on the order of 7000 USD per GPU-year of revenue that is about two basis points.

This is a stronger statement than the null it replaces, in three ways. It is positive rather than absent: the mechanism exists, binds, and has a measurable price. It is a claim about a distribution rather than a mean, so it does not evaporate under reaggregation. And it is robust to our sample restrictions, because it is a tail statistic — adding the withheld high-price zones can only increase it, and adding cheap zones such as ERCOT West contributes exactly zero.

5.4 The ordering is not what the annual means suggested

Ranking zones on the tail that governs curtailment, rather than on the mean, reorders them substantially. Belgium moves from fifth to first. Norway NO2 — which on annual means looked anomalously expensive for a nominally supply-rich zone, and which we previously reported as such — moves to *last*: it has the lowest maximum hourly cost in the sample and never goes uneconomic. On the statistic that matters, Norwegian hydro behaves exactly as the supply-rich classification predicts. Our earlier claim that the hub taxonomy fails was itself an artefact of the wrong statistic, and we withdraw it.

5.5 The dispersion is smaller than assumed

The thesis assumes roughly 3x variation. We measure 1.53x at zero opex, falling to 1.17x at 0.20 USD per GPU-hour.

This compression is not an artefact of our sample; it is algebraic. If power costs span $[p, 3p]$ and non-power opex is a common constant c , the shutdown-price ratio is

$$\frac{3p + c}{p + c}, \quad (6)$$

which equals 3 only when $c = 0$ and decreases monotonically in c . The headline “3x power costs imply 3x shutdown prices” inference is therefore false for any non-zero non-power operating cost, independently of data. Since plausible non-power opex is of the same order as — or larger than — the entire power cost, the dispersion that actually drives a merit order is materially below the dispersion in electricity prices that motivates it.

5.6 Break-even power prices, and the limits of a mean-based test

Inverting Equation (3) gives the electricity price at which a fleet sits exactly at its shutdown price:

$$p_{\text{kWh}}^* = \frac{\text{compute_price} - \text{opex}_{\text{marginal}}}{\text{delivered_kW}}. \quad (7)$$

At the cheapest compute price we observe — A100 spot, 0.86 USD per GPU-hour — the threshold is 0.740 USD kW⁻¹ h. At H100 on-demand list prices it is 12.80 USD kW⁻¹ h.

Compute price basis	USD/GPU-h	Break-even power (USD kW ⁻¹ h)
A100 spot	0.86	0.74
A100 reserved 3-year	1.59	1.40
H100 spot	2.61	2.33
H100 reserved 5-year	5.65	5.09
H100 on-demand list	14.13	12.80

Table 3: Electricity price required to force curtailment, at 0.05 USD/GPU-h non-power opex.

It is tempting to compare these thresholds against zone *mean* prices — the dearest zone in our sample averages 0.129 USD kW⁻¹ h, roughly a fifth of the lowest threshold — and conclude that the mechanism can never activate anywhere. An earlier draft of this paper did exactly that, and it was wrong.

Hourly prices breach these thresholds. Belgium’s dearest hour reaches 1.086 USD kW⁻¹ h, 47 % above the 0.740 USD kW⁻¹ h threshold; the Netherlands and Germany/Luxembourg also breach it. What the break-even calculation correctly establishes is therefore narrower than we first claimed: the mechanism cannot activate on *typical* prices in any zone we can foresee, so it will always be a tail phenomenon rather than a persistent cost ranking. How much that tail is worth is an empirical question, answered in Section 5 at roughly one dollar per GPU-year.

The general lesson is that a threshold test evaluated against a mean tells you nothing about a mechanism that operates in the tail. We record this because it is the error we made.

5.7 Why this cannot be rescued by better power data

The gap is not a measurement problem. Two comparisons from our own data make the scale clear. Azure’s list price for the same H100 SKU differs by 3.69 USD per GPU-hour between **eastus** and **westeurope** — a vendor pricing decision. The entire power-cost difference across our seven zones

is 0.05 USD per GPU-hour. The revenue leg’s regional dispersion is roughly 70x the cost leg’s, and it is driven by commercial policy rather than by electricity.

Capital cost sharpens the same point. An 8-accelerator H100 node at 300kUSD to 400kUSD over four to five years is on the order of 0.86 USD to 1.43 USD per GPU-hour of depreciation before any cost of capital — roughly 6 to 15x the power cost. We do not publish these figures in our dataset because we have no free, citable source for node prices, and they are stated here only to size the comparison. But the ordering is unambiguous: for any question other than short-run curtailment, power is not the term that matters.

6 Constraint analysis: chips versus power

[Pending: not tested. Interconnection-queue and grid-headroom data collection has not been carried out, so this paper offers no evidence on whether power or chip supply binds. We note only that the finding above is consistent with power mattering for siting and capital planning rather than for dispatch, since a cost that is one per cent of revenue can still be decisive when the alternative is not being able to energise a site at all. That is a different mechanism from the merit order and requires different data.]

7 Sensitivity

The conclusion is robust to every parameter we vary, because the result is not marginal — it is two orders of magnitude from the threshold.

Parameter	Range tested	Effect on headroom	Changes conclusion?
Non-power opex	0.00 USD to 0.20 USD/GPU-h	99.0 % → 97.6 %	No
Accelerator	A100 vs H100	96.7 % → 99.0 %	No
Zone	7 European zones	1.53 x cost spread	No
delivered_kW	0.9 kW to 1.4 kW	+55/−18% on a small item	No

Table 4: Sensitivity. To overturn the conclusion, compute prices would have to fall roughly 40-fold, or power costs rise roughly 40-fold; no combination of the parameters above comes close.

The honest way to state the fragility is this: the conclusion depends far more on the *level* of the compute price than on anything in the cost model. If the relevant compute price is not the hyperscaler list price but something closer to the spot price — roughly a fifth of list — headroom falls from 99 % to about 95 %. That is still not binding. Only a compute price near 0.15 USD per GPU-hour, some 95x below current list, would make the merit order operative.

8 Limitations

We list every material weakness rather than the flattering subset.

- **Seven zones, one continent, one partial year.** Licence restrictions removed five of twelve collected zones from the published sample, including Iberia and the Nordic zones that would most likely widen dispersion. Our 1.53x is therefore a lower bound on global dispersion, and the US, Gulf, China and India hubs are absent entirely.

- **No capacity weighting.** A merit-order curve should have installed capacity on its x-axis. We have no per-zone GPU capacity data, so Figure 2 shows the cost ladder without capacity weights. The curtailment *sequence* is unaffected; the quantity retiring at each step is unknown.
- **The compute-price leg is the weakest input.** No free redistributable index exists: Silicon Data’s SDH100RT and GetDeploying’s index are both paywalled, and the latter’s terms prohibit redistributing raw data. We rejected both under the free-data constraint rather than cite figures a reader cannot verify, and built our own basket instead — which is narrower than what we rejected.
- **One vendor, one segment.** The basket is Azure-only and hyperscaler-only. No neocloud provider is represented, so the segment dispersion we argue is central cannot yet be measured within a consistent methodology.
- **On-demand list price is not realised revenue.** Large operators transact on confidential multi-year reserved contracts at material discounts. This is the mirror of the PPA problem on the power side: both legs are public proxies for confidential transactions. Using list price *overstates* revenue and therefore overstates headroom.
- **Wholesale power price is not delivered power cost.** We compare a hyperscaler retail compute price against a wholesale commodity power price. A real operator pays network charges, levies and taxes on top of the wholesale price, which in several European jurisdictions exceed the commodity component. This *understates* power cost, possibly by more than a factor of two, and is the single most consequential methodological gap in this paper. Correcting it would raise the power share of revenue but would not plausibly close a hundredfold gap.
- **PUE and overhead are assumptions, not measurements.** Operator-reported PUE is self-reported and unaudited, and our 0.180kW overhead is a reconciling residual.
- **Capital cost is excluded by construction.** This is correct for a curtailment question and wrong for every other question one might ask. Our own rough sizing puts depreciation at 6 to 15x the power cost.
- **All power cost is treated as avoidable.** That is wrong for any operator on an industrial tariff with a demand charge, or under a take-or-pay PPA, where a large part of the power bill is incurred whether or not the machine runs. Correctly splitting avoidable from unavoidable power cost would *lower* the true shutdown price and make the merit order bind even less often than we report.
- **Claims 1 and 2 are in tension and we cannot resolve it.** If interconnection is genuinely the binding constraint, then grid capacity is contracted and largely pre-paid, and an operator holding a scarce interconnect runs it regardless of hourly energy price — which would suppress the very curtailment mechanism Claim 1 describes. The two claims can only both hold under specific contracting regimes, and we have not established which regime prevails.
- **Azure spot is not an auction clearing price.** In our data spot sits at a near-constant 18.5% of list across two accelerator generations and two continents, which is the signature of a published discount schedule rather than a market. We use it as a low revenue bound, not as evidence about where capacity clears.

- **List prices are stale relative to power prices.** Several price rows carry effective dates years old, while the power series is hourly. Any spread between a sticky annual list price and an hourly spot power price is dominated by the staleness of the former.

9 Conclusion

The compute merit order is real, computable from open data, and economically inert at current prices. Electricity is about one per cent of H100 rental revenue; headroom over avoidable cost exceeds 97 % everywhere we can measure; no zone is uneconomic; and the cross-zone dispersion in shutdown prices is 1.53x rather than the 3x assumed, compressing further once non-power operating costs are included.

The spark-spread analogy fails quantitatively. It works for gas generation because fuel is most of marginal cost, so fuel-price differences drive dispatch. It does not transfer to GPU compute, where power is a rounding error against rental revenue. Regional power-price differences of the magnitude observed cannot produce a curtailment sequence; they can only produce small margin differences.

This does not make power irrelevant to AI infrastructure — it relocates where power matters. Three candidate mechanisms survive our result and each requires different data than we assembled:

1. **Capital recovery, not dispatch.** If accelerator depreciation dominates full cost, the economically interesting threshold is the price at which new capacity stops being *built*, not the price at which existing capacity stops running. Our shutdown price is deliberately the wrong tool for that question.
2. **Availability, not price.** A one-per-cent cost item is still decisive when the alternative is being unable to energise a site. Interconnection queues and grid headroom plausibly bind on *whether* capacity exists rather than on what it costs to run.
3. **Compute price collapse.** If compute prices fell toward marginal cost, as they would in a commoditised market with excess capacity, the merit order would become operative. Our result quantifies how far that is: roughly a 40- to 95-fold decline.

We report this as a negative result because the data does not support the thesis as framed. The project’s next phase should redirect toward constraint data and capital cost, where the evidence suggests the binding economics actually lies.

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